

AI / ML Salaries



Presented by: Joma Chase

Why Show Interest?



The demand for AI talent is skyrocketing



Unlocking insights from data



Driving innovation and efficiency



Solving global challenges



Enhancing human capabilities

Project Goals



Discuss job popularity



Discuss position growth



Discuss the position with the highest pay



Discuss experience level with the best pay



Company size with the best pay

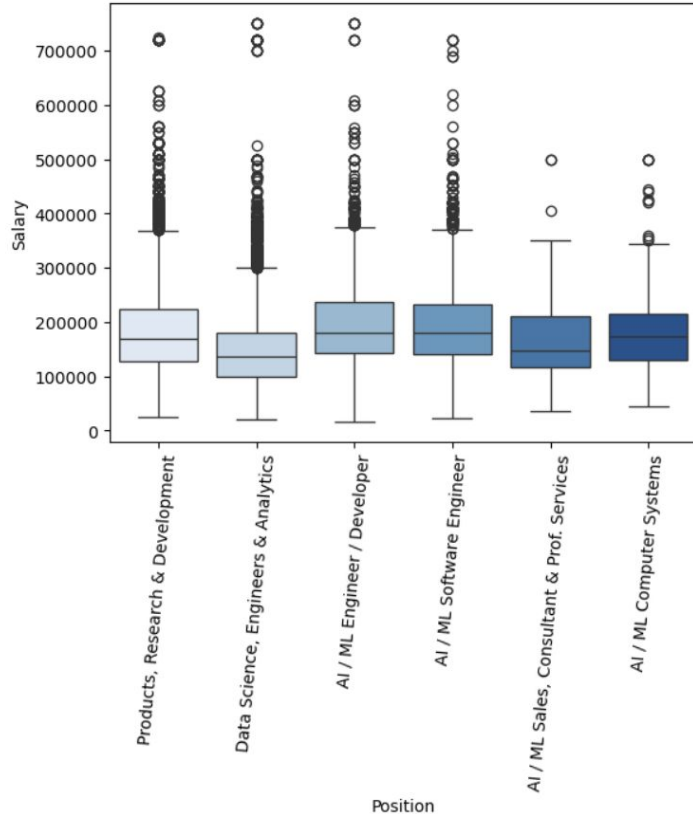
Data Collection

- Size : 73,148 to 65,877
- Year: 2020 - 2025
- The dataset was retrieved from GitHub (Cedric Aubin)
- Focus: Position, Experience_Level, Company_Size, Year and Salary

Data Cleaning

- Dropped columns (salary, employee_residence) to avoid duplicated data
- Checked for missing values, outliers, and duplicates
- Narrowed down job titles from 289 positions to 6 to simplify analysis using Excel
- Renamed column names and rows to make more readable

Position Focus:



- The job highest in demand is Data Science, Engineers and Analytics

Min salary: \$16, 000

Max salary: \$750, 000

Mean: \$145, 728

- Most highest salary is an AI/ML Engineer or Developer position

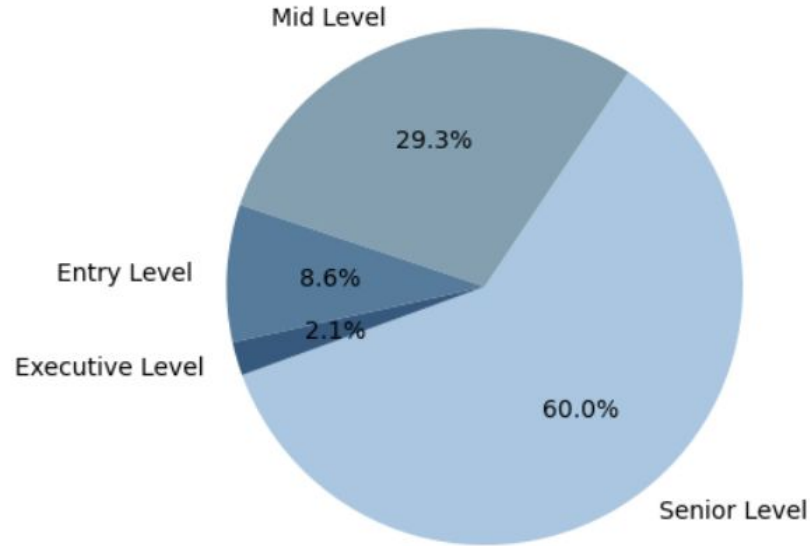
Mean salary: \$193, 610

p-value = 0

Null Hypothesis : reject

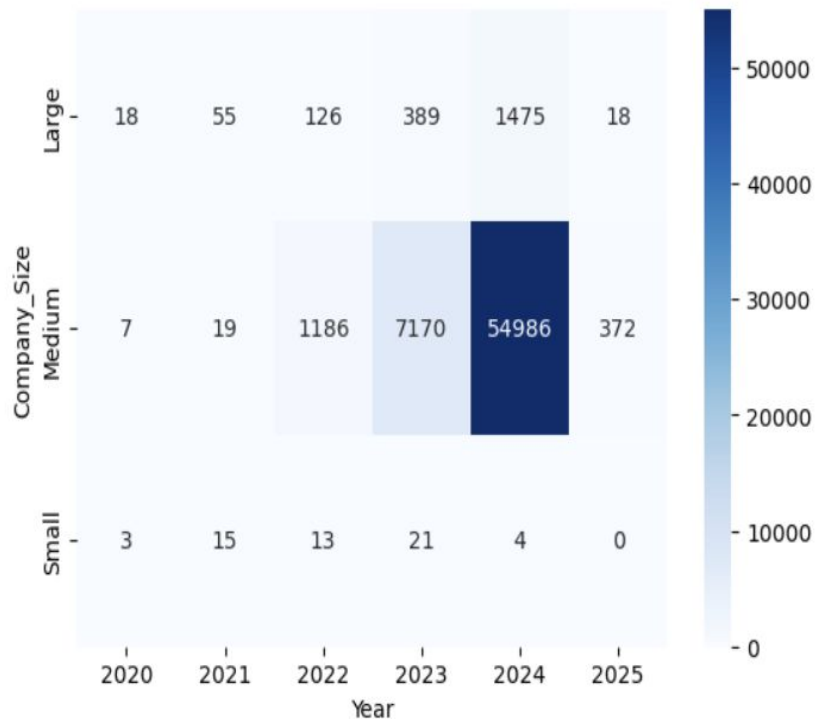
Experience Level Focus:

Distribution of Experience Levels



1. **Dominance of Senior Level:** The Senior Level category has the highest count at **39,552**.
2. **Considerable Mid-Level Representation:** With **19,280** entries in the Mid Level category, shows that there is a healthy representation of professionals
3. **Entry-Level Opportunities:** The Entry Level category, with **5,661** entries, suggests that there are opportunities available for newcomers to the workforce.
4. **Overall Distribution:** The distribution of experience levels shows a clear pyramid structure where the number of employees decreases as the level of experience increases.

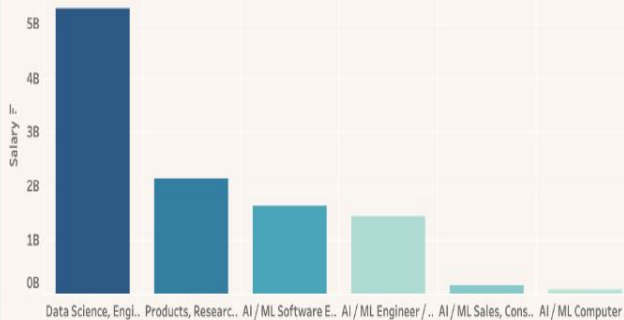
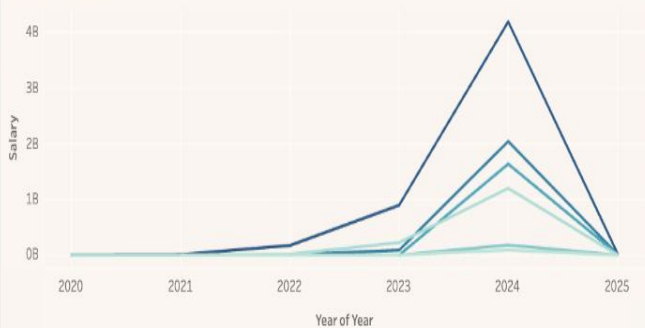
Company Size Focus:



Dominance of Medium-Sized Companies: The year 2024 shows a significant concentration of focus on medium-sized companies, with a striking value of 54,986.

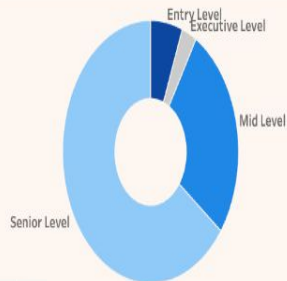
- Decline in Small Company Engagement:** The numbers for small companies are low across all years, especially in 2024.
- Gradual Growth in Large Company Engagement:** Although the values for large companies are lower compared to medium-size, there is a noticeable increase from 2020 to 2024. Particularly, the value of 1,475 in 2024.
- Fluctuations in Medium Company Engagement:** The values for medium companies fluctuate over the years although there is a spike after high of 7,170 in 2023.

Position Demand



Position According To Salary, Experience Level & Employment Type

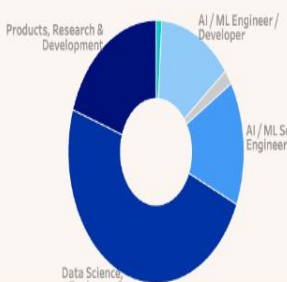
Experience Level with Best Salary



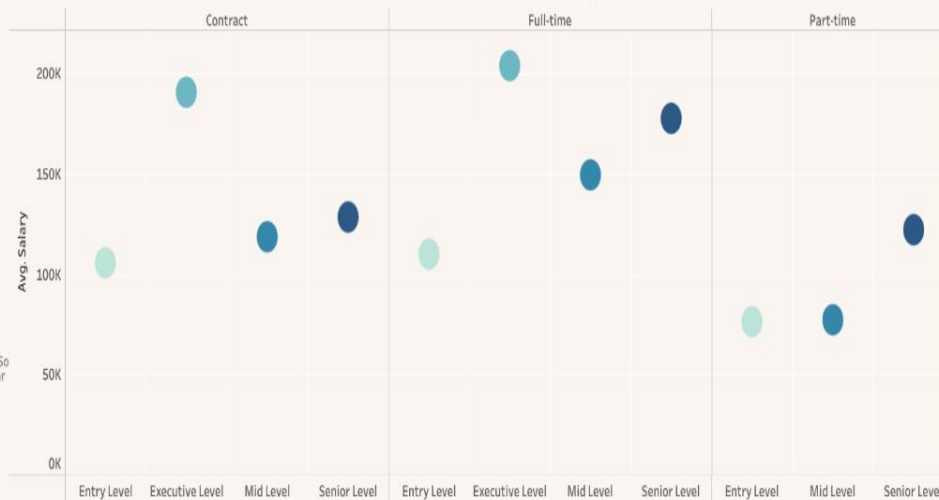
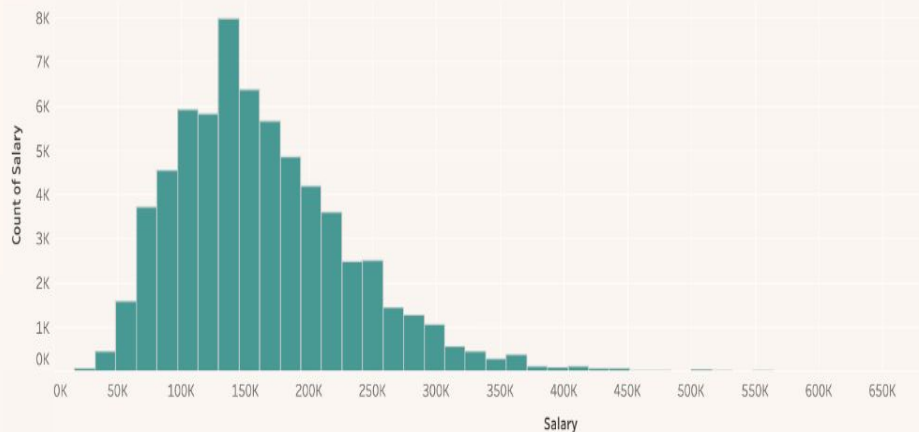
Company Size with Best Pay



Positions In Demand



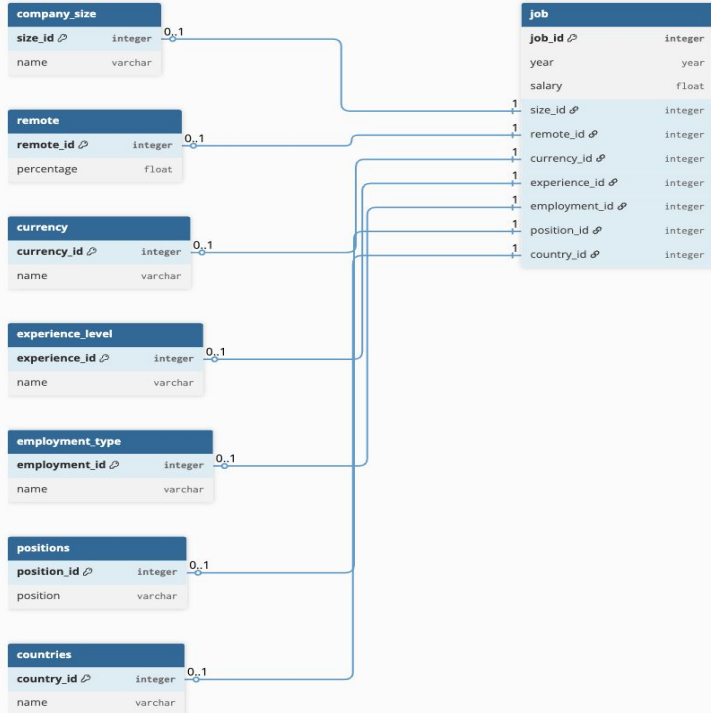
Salary Range



ERD Focus:

The ERD represents the database schema involving the relationships between the job and the entities that were analyzed.

- Total of 9 tables made up details about the model; company size, position names, employee experience, and salary compensation are stored and interrelated in SQL.
- SQL will further assist into manipulating what the data can enlighten us about these relationships



Delve Further With SQL:

Highest-paying positions in the US for remote jobs and their average salaries.

```
-- Listed the top 3 highest-paying positions in the US for remote jobs, and displayed their average salaries.
WITH RankedRemoteUSJobs AS (
  SELECT
    p.position,
    et.name AS employment_type,
    el.name AS experience_level,
    ROUND(AVG(j.salary), 2) AS average_salary,
    RANK() OVER (ORDER BY AVG(j.salary) DESC) AS salary_rank
  FROM
    job j
  JOIN
    positions p ON j.position_id = p.position_id
  JOIN
    countries c ON j.country_id = c.country_id
  JOIN
    remote r ON j.remote_id = r.remote_id
  JOIN
    employment_type et ON j.employment_id = et.employment_id
  JOIN
    experience_level el ON j.experience_id = el.experience_id
  WHERE
    c.name = 'US' AND r.percentage = 100
  GROUP BY
    p.position, et.name, el.name
)
```

CTE



Position	Employment Type	Employment Level	Average Salary
Data Science	Contract	Executive Level	\$ 41, 600
AI/ML Engineer	Full-time	Executive Level	\$ 240, 158
Products, R & D	Full-time	Executive Level	\$227, 392

Delve Further With SQL:

```
-- How does the average salary vary based on the company size for a particular year?  
• SELECT  
    cs.name AS company_size,  
    j.year,  
    ROUND(AVG(j.salary), 2) AS average_salary  
FROM  
    job j  
JOIN  
    company_size cs ON j.size_id = cs.size_id  
GROUP BY  
    cs.name,  
    j.year;
```



Join

How does the average salary vary based on the company size for a particular year?

	2025	2024	2023	2022
S	\$155, 000	\$130, 000	\$113, 000	\$134, 000
M	\$151, 000	\$163, 000	\$164, 000	\$152, 000
L	\$172, 000	\$157, 000	\$163, 000	\$152, 000

Delve Further With SQL:

```
-- Which positions have salaries higher than the overall average salary for the year 2025?  
SELECT position  
FROM positions  
WHERE position_id IN (  
    SELECT position_id  
    FROM job  
    WHERE year = 2025  
    AND salary > (SELECT AVG(salary) FROM job WHERE year = 2025)  
);
```

SubQ



1. Products, R & D
2. Data Science, Engineers, & Analytics
3. AI/ML Engineer / Developer
4. AI/ML Software Engineer
5. AI/ML Sales, Consultant & Prof. Services

Delve Further With SQL:

Positions with Highest Salary Based on Experience Level

```
-- Create a temporary table of positions with the highest salaries for each experience level
• DROP TEMPORARY TABLE TempHighestSalaries;
• CREATE TEMPORARY TABLE IF NOT EXISTS TempHighestSalaries AS
WITH RankedSalaries AS (
    SELECT p.position, el.name AS experience_level, j.salary,
           ROW_NUMBER() OVER (PARTITION BY el.name ORDER BY j.salary DESC) AS salary_rank
    FROM job j
    JOIN positions p ON j.position_id = p.position_id
    JOIN experience_level el ON j.experience_id = el.experience_id
)

SELECT position, experience_level, salary
FROM RankedSalaries
WHERE salary_rank = 1;

• SELECT * FROM TempHighestSalaries;
```

**Temp
Table**



Position	Experience Level	Salary
Data Science	Senior Level	\$ 750, 000
Data Science	Mid Level	\$750, 000
Data Science	Entry level	\$ 720,000
Products, R & D	Executive Level	\$ 609, 000

The Big Picture: Key AI/ML Market Takeaways



- Data Science, Engineering, and Analytics and Products, Research & Development roles are leading the demand across the market
- Top Earners: AI/ML Engineers & Developers lead the pay charts, with mean salaries reaching \$193,610 and \$189,431, respectively.
- Compensation King: Senior-level roles consistently command the highest compensation packages.
- 2024 emerged as a breakout year for AI and ML hiring momentum
- Best Value: Medium-sized companies are offering the most competitive overall salary packages.

Lessons Learned & Next Steps for Deeper Analysis

Data Refinement Opportunities:

- Data Normalization: Extensive cleaning was required
- Global Context: Future analysis requires broader global data coverage for a complete picture.
- Temporal Granularity: Adding month-to-month data will allow for tracking seasonal/short-term hiring trends.
- Regional Deep Dive: A state-by-state analysis is necessary for regional compensation comparisons.
- Data Health Flagged: Noted a potential experience-level imbalance and flagged extreme salary outliers for targeted investigation in future work.
- Ran into outliers - Flag for analysis

Technical Takeaways:

- Tool Flexibility: Successfully navigated data cleaning hurdles by leveraging Excel after initial Python challenges.
- SQL Optimization: Complex joins required significant effort; future work should prioritize SQL proficiency for database querying efficiency.

Thank You!



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